Sora Kim

Regulatory HCI · STS · Computational Methods · Accountability in Regulated Systems

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RESEARCH FOCUS

Grounded in four years of pharmaceutical Good Manufacturing Practice (GMP) field experience, I study how accountability converges prematurely in regulated organizations — how responsibility attribution settles on fast, documentable explanations before alternative causal accounts have been adequately examined. I develop this through three linked components: theoretical frameworks for responsibility convergence (AAC–CF–RC–PAC lifecycle), prototype-based Human-Computer Interaction (HCI) interventions that preserve comparison in regulated investigation workflows, and computational modeling of accountability concentration in regulatory documents. My goal is to design accountability-aware AI systems for regulated environments: systems that preserve the conditions under which genuine human judgment is possible, without altering regulatory constraints.

Interests: Regulatory HCI · Science and Technology Studies (STS) · computational methods · accountability · regulated organizations · AI-mediated reasoning · judgment-preserving design

RESEARCH PROGRAM

Theoretical contributions Anticipated Accountability Convergence (AAC) – Cognitive Friction (CF) – Responsibility Convergence (RC) – Premature Accountability Convergence (PAC) lifecycle: a four-stage account of how responsibility is prospectively aligned before action, procedurally embedded during execution, retrospectively stabilized under closure pressure, and accelerated through AI-mediated representation.

Computational validation FDA Warning Letter corpus (N=50, 160 violations, 2016–2025): PAC detection AUC $>$ 0.97; accountability graph star topology (Firm in-degree centrality = 0.444); responsibility attribution macro F1 = 0.725 (Individual AUC = 0.940); structural pre-convergence — 159/160 violations converge to Firm attribution by document architecture before investigative reasoning begins.

Empirical findings across four HCI studies (N=20 GMP-experienced participants each):

- *Accessibility layer*: keyword-based retrieval reduced exploratory Standard Operating Procedure (SOP) search time from ≈6 min to ≈1 min; access constraints restrict behavioral repertoire — practitioners act within personally known SOP subsets rather than most applicable procedures.
- *Structural layer*: loopback behavior observed across all 20 sessions (mean ≈5/session); less experienced practitioners averaged 7–10 loopbacks vs. 2–3 for experienced — not a reasoning gap but a design gap.
- *Coordination layer*: all 20 participants anticipated reduced accountability asymmetry from friction visibility; checklist vs. pattern detection dissociation demonstrates that usability and immediate legibility are not the same thing.
- *Reasoning layer*: 15/20 produced single-cause explanations under closure pressure (confidence avg. 1/5). AI scaffolding shifted 12/15 from person-centered to systemic attribution; usability 5.0/5.

SELECTED PUBLICATIONS

- STS **Responsibility Convergence in Organizational Investigations**
  SSRN Preprint, 2026
  Introduces responsibility convergence as a structural property of organizational investigations: under closure

pressure, attribution systematically narrows toward fast, documentable explanations before epistemic work is done.

- STS **Premature Accountability Convergence in Regulated Organizations**
  SSRN Preprint, 2026
  Introduces PAC as a distinct failure mode: AI representations accelerate pre-existing convergence tendencies, compressing interpretive space before alternative accounts remain viable.
- Comp **Responsibility as a Multi-Layer Signal: A Unified Computational Framework for Accountability Convergence in Regulatory Investigations**
  arXiv Preprint, 2026
  Integrates PAC detection (AUC $>$ 0.97), accountability graph extraction (Firm in-degree centrality = 0.444), and responsibility attribution prediction (macro F1 = 0.725) into a unified framework. Key finding: FDA enforcement documents exhibit structural pre-convergence — accountability is encoded in document architecture before investigative reasoning begins.
- HCI **Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations**
  SSRN Preprint, 2026
  Reframes premature causal convergence as an interaction problem; introduces comparison preservation as a design principle. AI as comparison enforcer broadens causal accounts (18/20) and shifts attribution from person-centered to systemic explanations (12/15). Usability 5.0/5; adoption intent unanimous.

WORKING PAPERS

- STS **Anticipated Accountability Convergence: Structuring Responsibility Before Action in Regulated Organizations**
  SSRN Preprint, 2026
  Process-level framework explaining how procedural, documentary, and evaluative infrastructures align responsibility expectations prospectively, before action unfolds.
- STS **Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts**
  SSRN Preprint, 2026
  Introduces cognitive friction as a document-level property; rule-based computational annotation of 500 procedural sentences; conditional branching identified as primary structural anchor of high-friction configurations.
- HCI **How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design**
  SSRN Preprint, 2026
  Introduces Regulatory HCI as a framework reconstituting regulated environments as interaction systems. Four layers (accessibility, structure, coordination, reasoning) with three design principles: upstream stabilization, observability over optimization, judgment preservation.
- HCI **When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation**
  SSRN Preprint, 2026
  Document traceability as an HCI problem; loopback reframed as structural signal rather than usability failure. Less experienced practitioners averaged 7–10 loopbacks vs. 2–3 for experienced — a design gap, not a reasoning gap.
- HCI **From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work**
  SSRN Preprint, 2026
  Access constraints restrict behavioral repertoire — practitioners act within personally known SOP subsets. Retrieval time reduced from $\approx$6 min to $\approx$1 min. N=20; adoption intent unanimous.
- HCI **Friction Before Failure: Making Review Burden Visible in Regulated Document Work**

SSRN Preprint, 2026

Workflow dashboard and TF-IDF pattern detection layer surfaces hidden coordination burden. Central finding: dissociation between immediate legibility and final endorsement. All 20 participants anticipated reduced accountability asymmetry.

CONFERENCE PRESENTATIONS

- HCI **Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure**
  Poster · CSCW 2026 · Under review
- STS **Premature Accountability Convergence: Closure Pressure, Regulatory Documentation, and Counter-Convergence Design in Pharmaceutical Investigations**
  4S Toronto 2026 · Under review

PROFESSIONAL EXPERIENCE

Quality Control / Quality Systems Project Lead Sep 2022 – Present

Pharmaceutical Manufacturing Company · South Korea

- Managed 20+ deviation and Corrective and Preventive Action (CAPA) investigations end-to-end in GMP-regulated pharmaceutical manufacturing — directly observing how documentation architecture shapes which causal explanations survive institutional pressure.
- Supported 10+ regulatory inspections and global audits; developed working expertise in how traceability, coordination, and responsibility attribution are structured through document design.
- Revised and maintained 50+ SOPs across validation, deviation, CAPA, and Root Cause Analysis (RCA) workflows; the gap between what procedures prescribe and how they are interpreted under time pressure motivates the cognitive friction and convergence work.
- Designed the organization's first documentation framework for pharmaceutical technology transfer — establishing accountability structures where none previously existed.

METHODS & SKILLS

Qualitative	Think-aloud protocol; qualitative coding (open, axial); document and corpus analysis; annotation protocol design; causal reasoning analysis
Quantitative	Experimental design (within-subject, task-based); behavioral logging; NLP / text processing (TF-IDF, sentence-level annotation); machine learning (classification, LOO-CV, ablation, SHAP); graph analysis (centrality, clustering); composite index construction (PCA, Ridge Regression)
Tools	Python · R · Streamlit · LaTeX · HTML/CSS · GitHub
Languages	Korean (native) · English (fluent) · Japanese (fluent)